

Exploratory Data Analysis (EDA) Report – Marketing Campaign Analytics

1. Introduction

This report presents an exploratory data analysis (EDA) of the marketing campaign dataset using Python. The primary objective of this analysis is to examine customer demographics, purchasing behavior, and campaign response patterns in order to uncover actionable insights that support retail analytics, customer segmentation, and data-driven marketing decision-making.

2. General Data Overview

2.1 Dataset Preview and Libraries Loading

The required Python libraries for data analysis and visualization were imported prior to conducting the exploratory analysis. The first few rows of the dataset were then displayed using `df.head()` to inspect the dataset structure, understand the available variables, and verify that the dataset was loaded successfully.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as mtick
import seaborn as sns
import plotly.express as px
from bokeh.plotting import figure, show
from bokeh.io import output_notebook
from scipy import stats
from scipy.stats import chi2_contingency
import sqlite3
import re

# Read the csv file and store it as a DataFrame named df
df = pd.read_csv('marketing_campaign_clean.csv')
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	...	NumCatalogPurchases	NumStorePurchases	NumWebVisitsMonth	AcceptedCmp3	AcceptedCmp4	AcceptedCmp5	AcceptedCmp1	Ac
0	7734	1993	Graduation	Single	79244	0	0	19-12-2012	58	471	...	10	7	1	0	0	1	1	
1	4369	1957	Master	Single	65487	0	0	10/1/2014	48	240	...	5	6	2	0	0	0	0	
2	433	1958	Master	Single	61331	1	1	10/3/2013	42	534	...	1	6	8	0	0	0	0	
3	7660	1973	PhD	Single	35860	1	1	19-05-2014	37	15	...	1	2	5	1	0	0	0	
4	92	1988	Graduation	Single	34176	1	0	12/5/2014	12	5	...	0	4	6	0	0	0	0	

5 rows × 28 columns

2.2 Dataset Dimensions

The dimensions of the dataset were examined using the dataset shape to determine the total number of observations (rows) and variables (columns), providing an overview of the dataset size and analytical scope.

```
# Check the dataframe Shape
print("Dimensions (rows, columns):",df.shape)

Dimensions (rows, columns): (2240, 28)
```

2.3 Data Types Inspection and Missing Values Analysis

The data types of all variables were reviewed using `df.dtypes` to ensure that each column was stored in an appropriate format for analysis. Additionally, missing values were identified using `df.isnull().sum()` to assess data completeness and detect any potential data quality issues prior to performing further analysis.

# Check for any missing values in the dataset <code>df.isnull().sum()</code>	# Check the data types of each column <code>df.dtypes</code>
ID 0	ID int64
Year_Birth 0	Year_Birth int64
Education 0	Education object
Marital_Status 0	Marital_Status object
Income 0	Income int64
Kidhome 0	Kidhome int64
Teenhome 0	Teenhome int64
Dt_Customer 0	Dt_Customer object
Recency 0	Recency int64
MntWines 0	MntWines int64
MntFruits 0	MntFruits int64
MntMeatProducts 0	MntMeatProducts int64
MntFishProducts 0	MntFishProducts int64
MntSweetProducts 0	MntSweetProducts int64
MntGoldProds 0	MntGoldProds int64
Total_spending 0	Total_spending int64
NumDealsPurchases 0	NumDealsPurchases int64
NumWebPurchases 0	NumWebPurchases int64
NumCatalogPurchases 0	NumCatalogPurchases int64
NumStorePurchases 0	NumStorePurchases int64
NumWebVisitsMonth 0	NumWebVisitsMonth int64
AcceptedCmp3 0	AcceptedCmp3 int64
AcceptedCmp4 0	AcceptedCmp4 int64
AcceptedCmp5 0	AcceptedCmp5 int64
AcceptedCmp1 0	AcceptedCmp1 int64
AcceptedCmp2 0	AcceptedCmp2 int64
Complain 0	Complain int64
Response 0	Response int64
dtype: int64	dtype: object

3. Descriptive Statistics

3.1 Summary Statistics

Summary statistics were computed to understand the central tendency and dispersion of the dataset. Key measures such as **mean**, **median**, **minimum**, **maximum**, and **standard deviation** were calculated for all numerical variables.

- **Option A:** The `describe()` function was used to generate an overall statistical summary of the dataset, providing a quick overview of all numerical features.
- **Option B:** Median values were additionally calculated separately to complement the `describe()` output, ensuring a more robust understanding of the data distribution, especially in the presence of potential outliers.

These two approaches together provide a comprehensive view of the dataset's distribution and variability.

Option A

	ID	Year_Birth	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts	...	NumCatalogPurchases	NumStorePurchases	NumWebVisitsMonth	AcceptedCmp3	AcceptedCmp4
count	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	...	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000
mean	5592.159821	1968.805804	51473.545982	0.444196	0.506250	49.109375	303.935714	26.302232	166.950000	37.525446	...	2.662054	5.790179	5.316518	0.072768	0.074554
std	3246.662198	11.984069	21909.882797	0.538398	0.544538	28.962453	336.597393	39.773434	225.715373	54.628979	...	2.923101	3.250958	2.426645	0.259813	0.262728
min	0.000000	1893.000000	1730.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	...	0.000000	0.000000	0.000000	0.000000	0.000000
25%	2828.250000	1959.000000	34686.250000	0.000000	0.000000	24.000000	23.750000	1.000000	16.000000	3.000000	...	0.000000	3.000000	3.000000	0.000000	0.000000
50%	5458.500000	1970.000000	51025.500000	0.000000	0.000000	49.000000	173.500000	8.000000	67.000000	12.000000	...	2.000000	5.000000	6.000000	0.000000	0.000000
75%	8427.750000	1977.000000	68275.750000	1.000000	1.000000	74.000000	504.250000	33.000000	232.000000	50.000000	...	4.000000	8.000000	7.000000	0.000000	0.000000
max	11191.000000	1996.000000	162397.000000	2.000000	2.000000	99.000000	1493.000000	199.000000	1725.000000	259.000000	...	28.000000	13.000000	20.000000	1.000000	1.000000

8 rows × 25 columns

Option B

```
# summary statistics for numerical columns (mean, median, min, max, and standard deviation) option B
summary = df.describe().T
summary["median"] = df.median(numeric_only=True)

summary = summary[["mean", "median", "min", "max", "std"]]
print(summary)
```

	mean	median	min	max	std
ID	5592.159821	5458.5	0.0	11191.0	3246.662198
Year_Birth	1968.805804	1970.0	1893.0	1996.0	11.984069
Income	51473.545982	51025.5	1730.0	162397.0	21909.882797
Kidhome	0.444196	0.0	0.0	2.0	0.538398
Teenhome	0.506250	0.0	0.0	2.0	0.544538
Recency	49.109375	49.0	0.0	99.0	28.962453
MntWines	303.935714	173.5	0.0	1493.0	336.597393
MntFruits	26.302232	8.0	0.0	199.0	39.773434
MntMeatProducts	166.950000	67.0	0.0	1725.0	225.715373
MntFishProducts	37.525446	12.0	0.0	259.0	54.628979
MntSweetProducts	27.062946	8.0	0.0	263.0	41.280498
MntGoldProds	44.021875	24.0	0.0	362.0	52.167439
Total_spending	605.798214	396.0	5.0	2525.0	602.249288
NumDealsPurchases	2.325000	2.0	0.0	15.0	1.932238
NumWebPurchases	4.084821	4.0	0.0	27.0	2.778714
NumCatalogPurchases	2.662054	2.0	0.0	28.0	2.923101
NumStorePurchases	5.790179	5.0	0.0	13.0	3.250958
NumWebVisitsMonth	5.316518	6.0	0.0	20.0	2.426645
AcceptedCmp3	0.072768	0.0	0.0	1.0	0.259813
AcceptedCmp4	0.074554	0.0	0.0	1.0	0.262728
AcceptedCmp5	0.072768	0.0	0.0	1.0	0.259813
AcceptedCmp1	0.064286	0.0	0.0	1.0	0.245316
AcceptedCmp2	0.013393	0.0	0.0	1.0	0.114976
Complain	0.009375	0.0	0.0	1.0	0.096391
Response	0.149107	0.0	0.0	1.0	0.356274

3.2 Initial Distribution Analysis

To understand the distribution patterns of key variables, **histograms** were generated for **Income** and **Year of Birth**.

These visualizations helped in:

- Identifying the **shape of the distribution** (normal, skewed, etc.)
- Observing the **spread and concentration** of values
- Detecting any initial signs of **anomalies or irregular distributions**

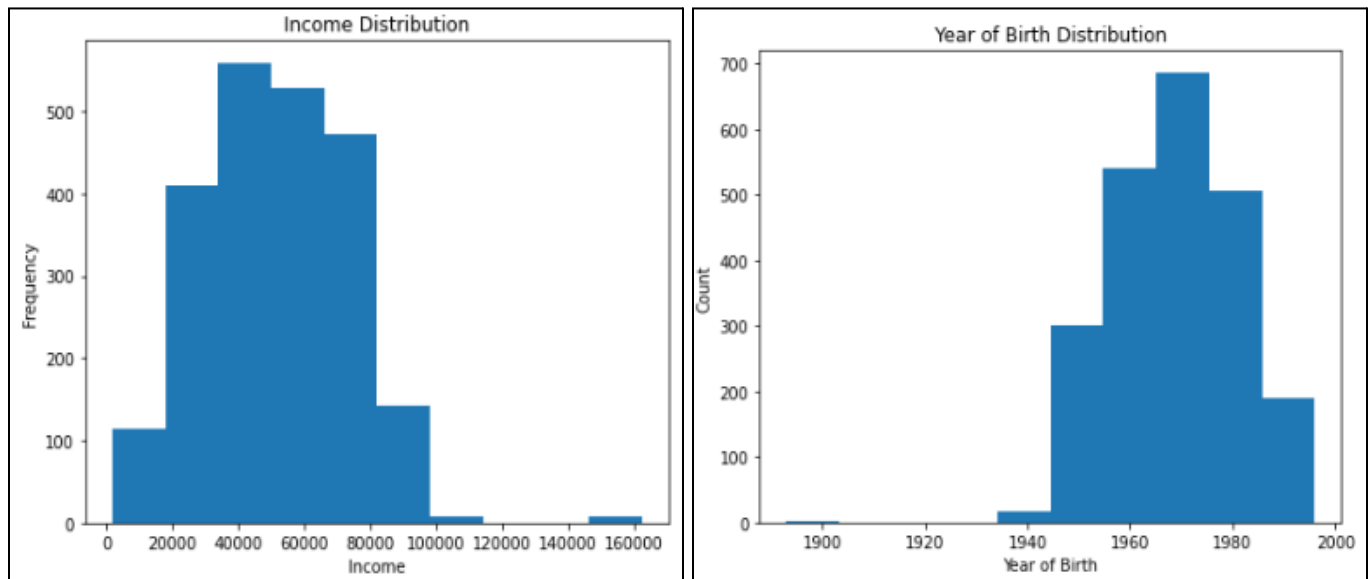
Overall, the histograms provided an initial understanding of how the data is distributed across key numerical features.

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Income histogram
axes[0].hist(df["Income"], bins=10)
axes[0].set_title("Income Distribution")
axes[0].set_xlabel("Income")
axes[0].set_ylabel("Frequency")

# Year_Birth histogram
axes[1].hist(df["Year_Birth"], bins=10)
axes[1].set_title("Year of Birth Distribution")
axes[1].set_xlabel("Year of Birth")
axes[1].set_ylabel("Count")

plt.tight_layout()
plt.show()
```



3.3 Initial Outlier Detection

To detect unusual or extreme values, **boxplots** were created for **Income** and **Recency**.

These plots were used to:

- Identify **outliers outside the interquartile range (IQR)**
- Visualize the **data spread and central clustering**
- Highlight potential data quality issues or extreme customer behaviors

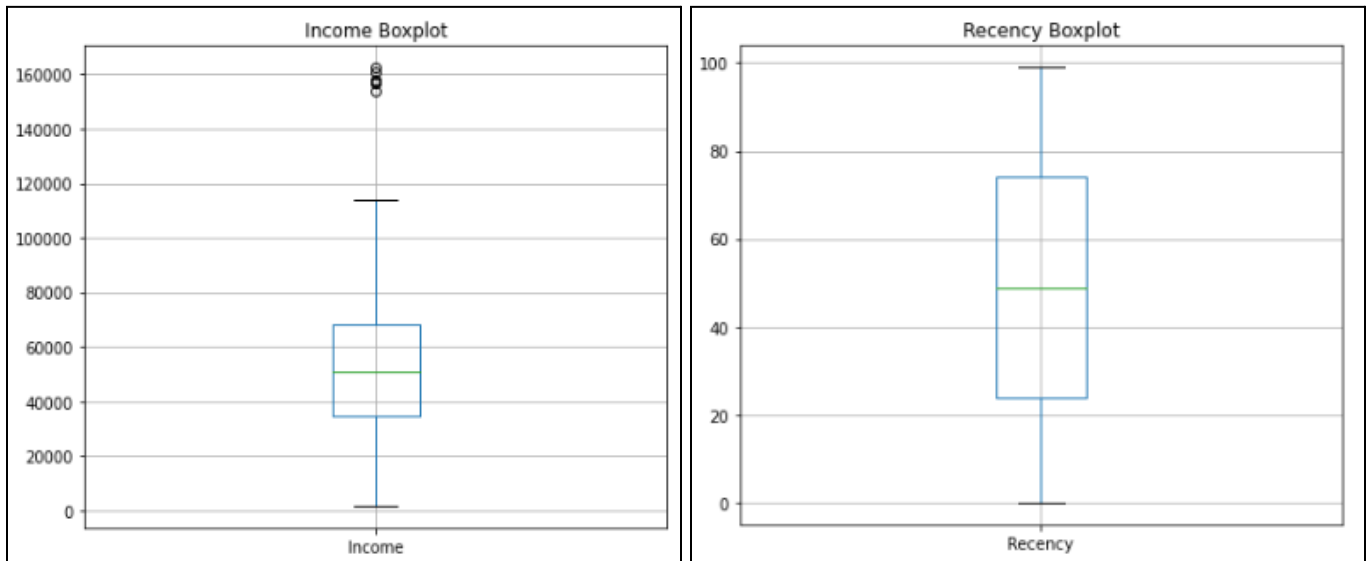
The boxplots indicated whether further data cleaning or transformation may be required before deeper analysis.

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Income boxplot
df.boxplot(column="Income", ax=axes[0])
axes[0].set_title("Income Boxplot")

# Recency boxplot
df.boxplot(column="Recency", ax=axes[1])
axes[1].set_title("Recency Boxplot")

plt.tight_layout()
plt.show()
```



4. Feature Engineering

4.1 Age Calculation

A new feature, **Age**, was created to enable age-based customer analysis. This was derived from the existing **Year_Birth** column by subtracting it from the current year:

$\text{Age} = 2026 - \text{Year_Birth}$

This transformation allowed the dataset to represent customers in terms of their current age, making it easier to analyze behavioral patterns and trends across different age groups.

```
df['Age'] = 2026 - df['Year_Birth']
```

5. Univariate Analysis

5.1 Numerical Variable Distributions

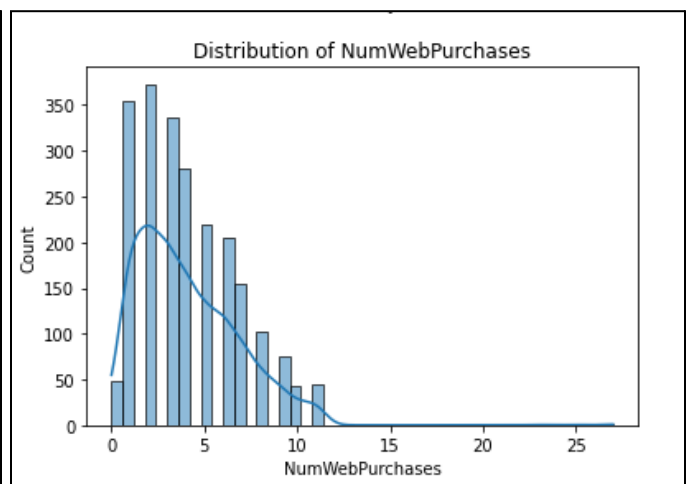
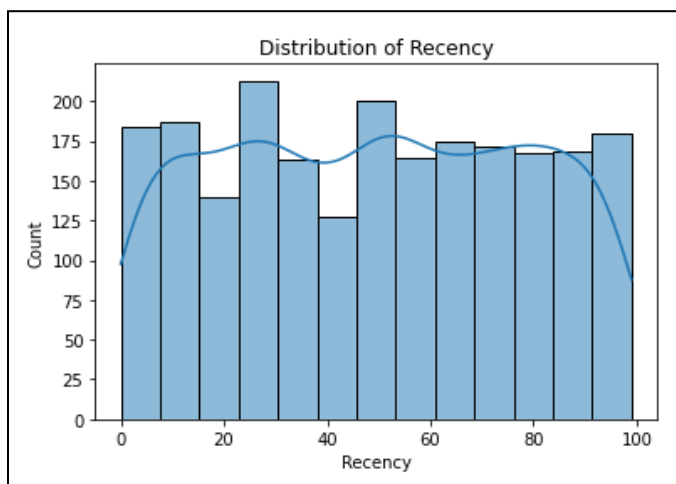
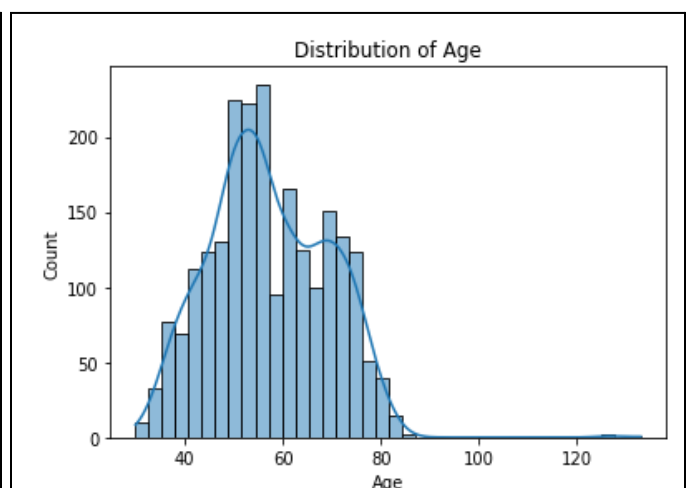
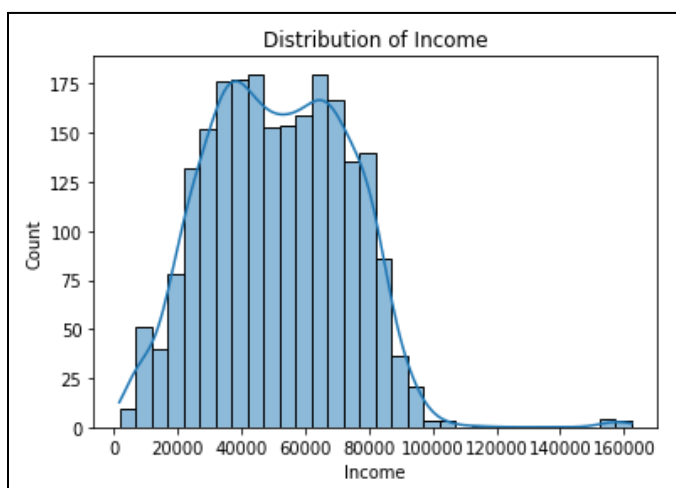
To understand the distribution of numerical features, **histograms with Kernel Density Estimate (KDE) curves** were generated. These visualizations were used to analyze the shape, spread, and skewness of key variables in the dataset.

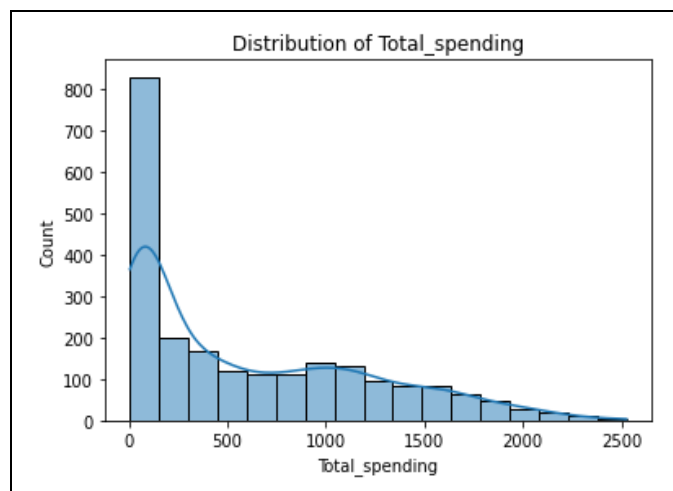
This helped in identifying:

- Whether the variables follow a normal or skewed distribution
- The concentration of values within specific ranges
- Any visible patterns or anomalies in the data

```
selected_num_cols = ['Income', 'Age', 'Recency', 'NumWebPurchases', 'Total_spending']

# Numerical Variables → Histograms
for col in selected_num_cols:
    plt.figure()
    sns.histplot(df[col], kde=True)
    plt.title(f'Distribution of {col}')
    plt.show()
```





5.2 Categorical Variable Distributions

To explore the structure of categorical features, **bar charts** were created for **Education** and **Marital Status**.

These visualizations provided insights into:

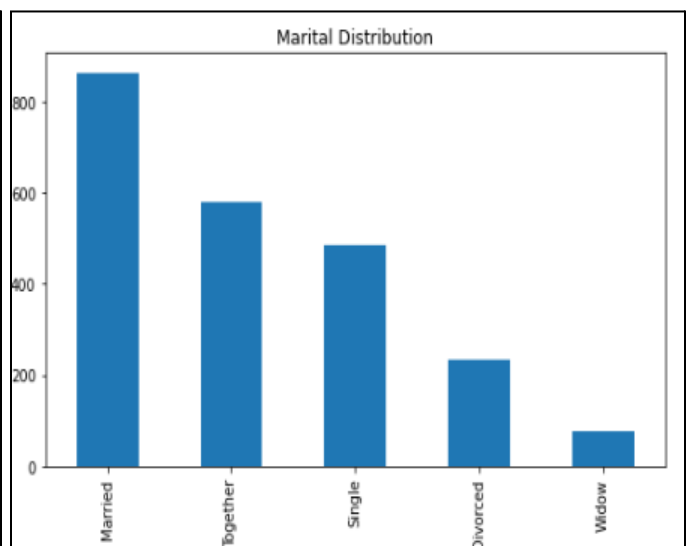
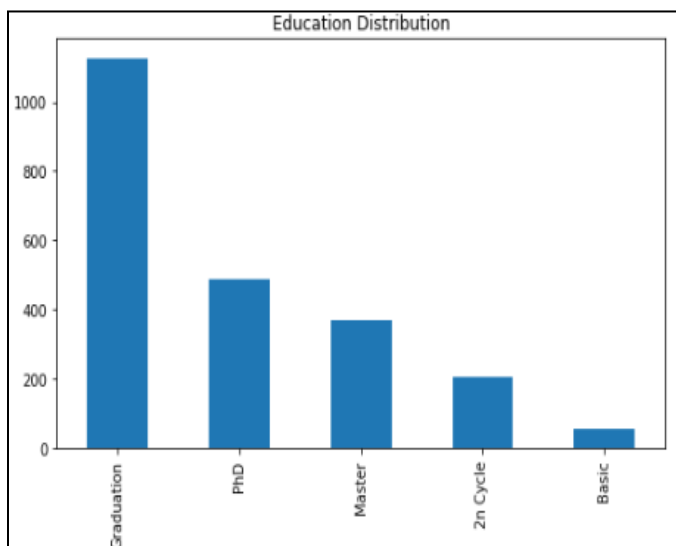
- The distribution of customers across different education levels
- The frequency of each marital status category
- Potential dominance or imbalance within categorical variables

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Education Distribution
df['Education'].value_counts().plot(
    kind='bar',
    ax=axes[0],
    title='Education Distribution')

# Marital Status Distribution
df['Marital_Status'].value_counts().plot(
    kind='bar',
    ax=axes[1],
    title='Marital Status Distribution')

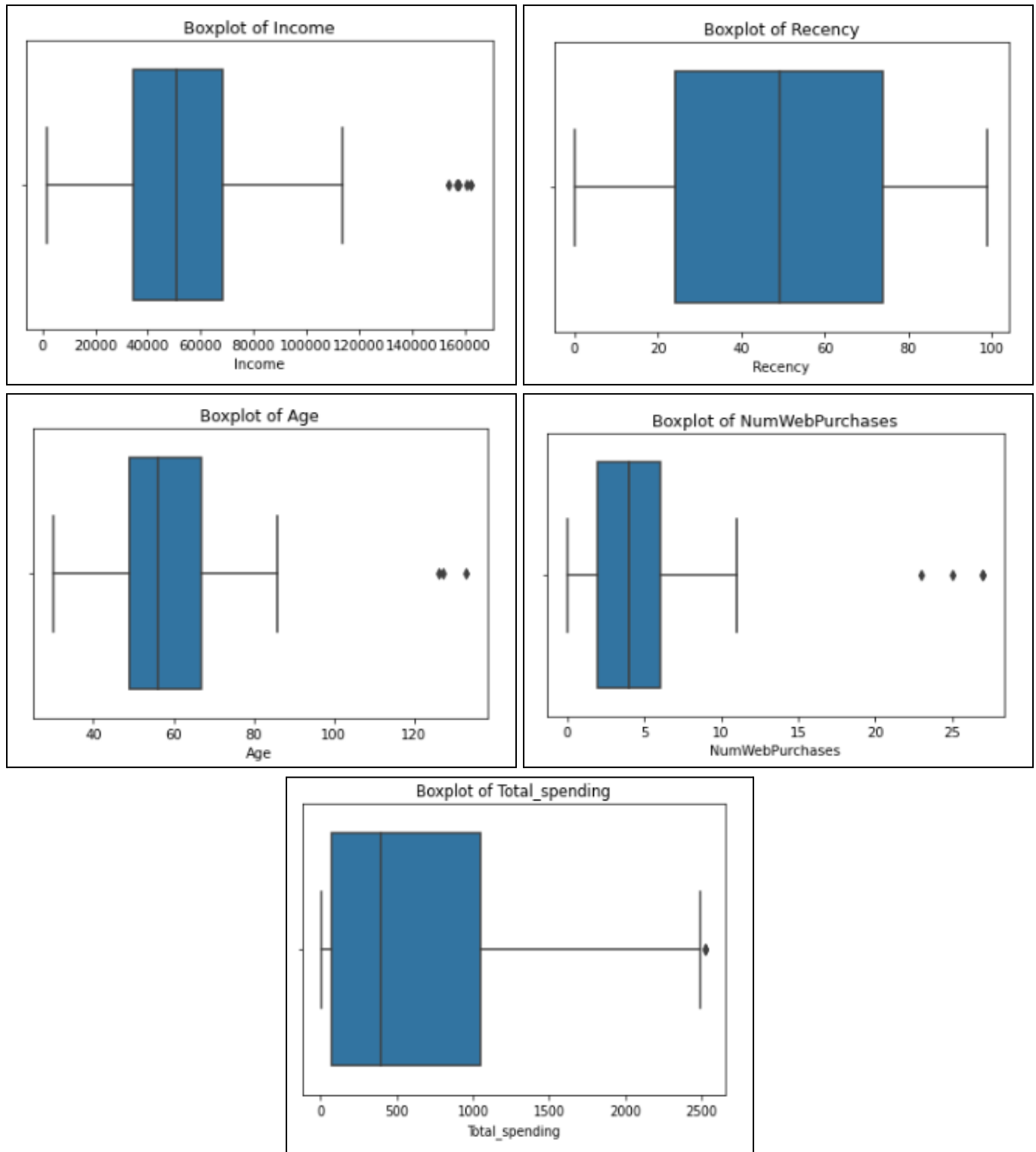
plt.tight_layout()
plt.show()
```



5.3 Outlier Analysis

To detect extreme or unusual values, **boxplots** were generated for all selected numerical variables.

```
# Outliers \ Boxplots
for col in selected_num_cols:
    plt.figure()
    sns.boxplot(x=df[col])
    plt.title(f'Boxplot of {col}')
    plt.show()
```



6. Bivariate Analysis

6.1 Numerical Variables vs Campaign Response

To examine relationships between customer characteristics and campaign response, **scatter plots** and **boxplots** were used to compare numerical variables against the **campaign response** variable.

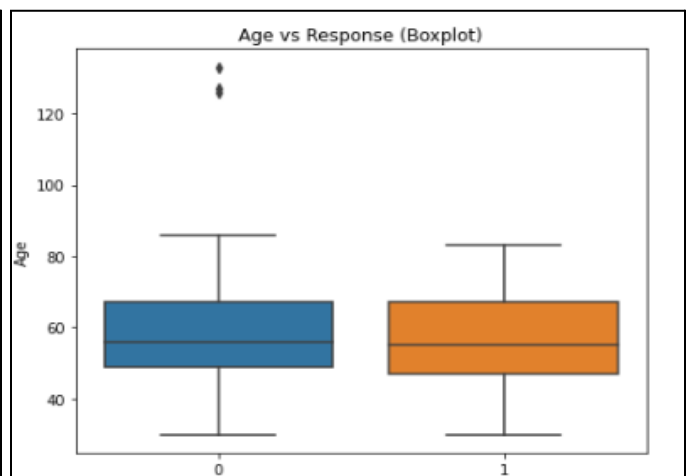
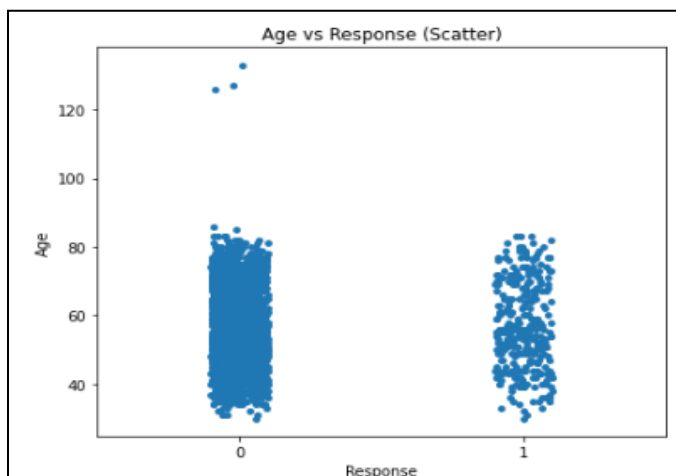
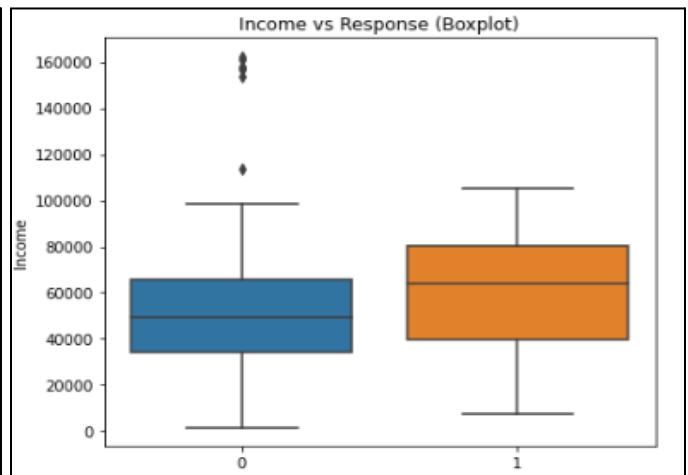
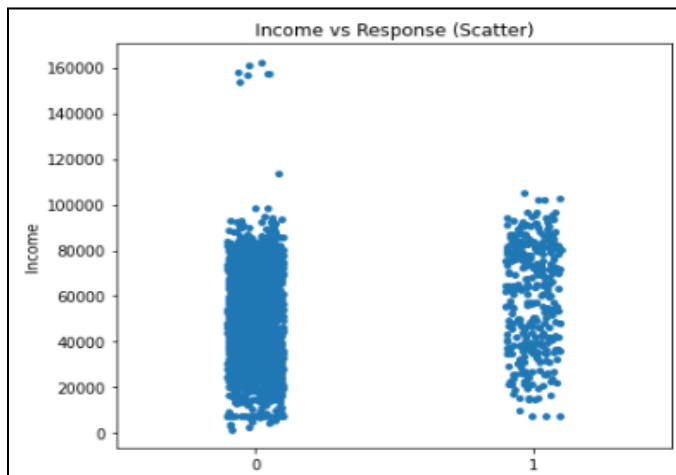
This analysis helps in understanding whether factors such as **income, age, and purchasing behavior** have any influence on a customer's likelihood to respond to marketing campaigns. It also supports the identification of patterns or trends that may explain differences in customer engagement levels.

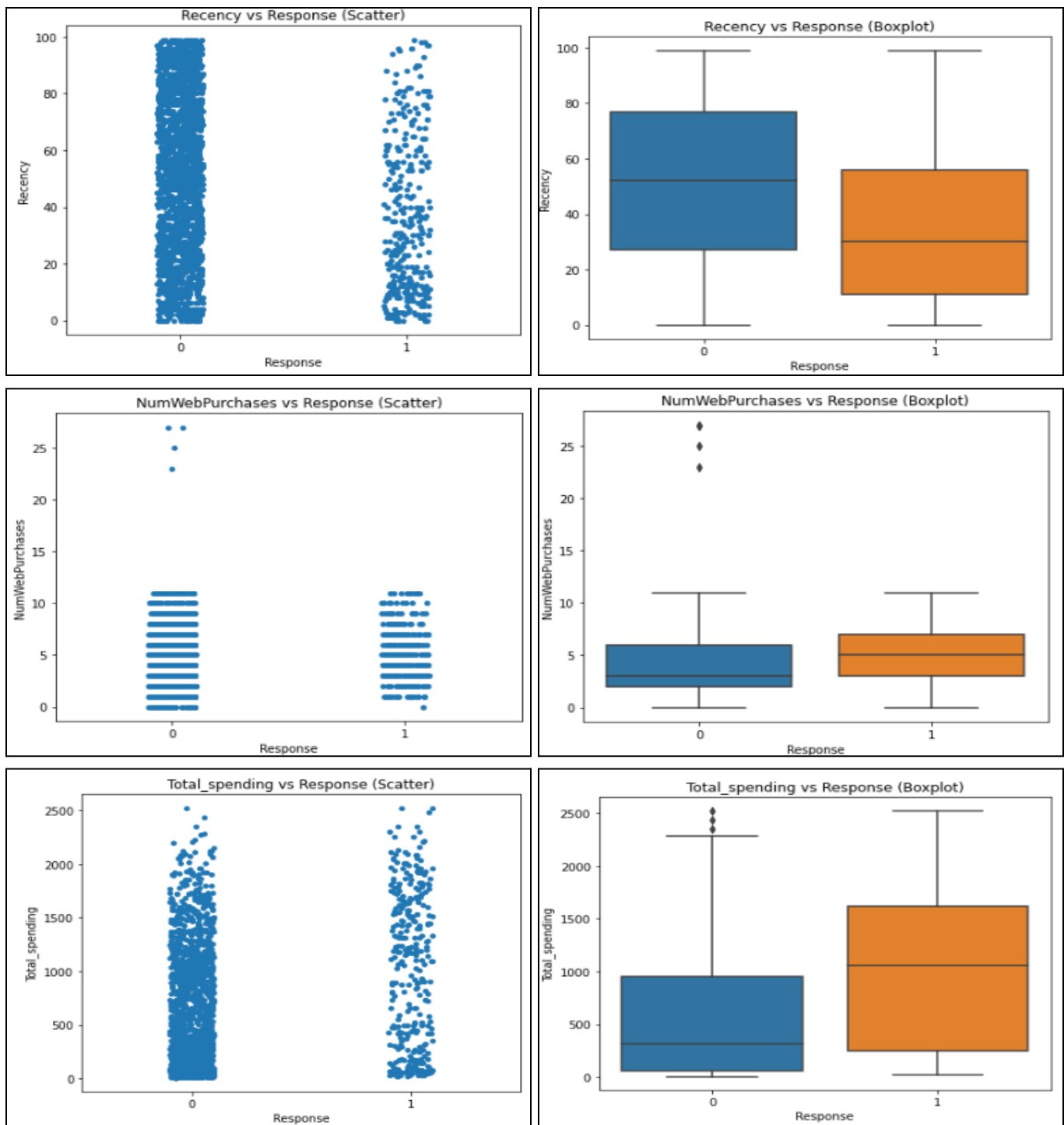
```
for col in selected_num_cols:
    fig, axes = plt.subplots(1, 2, figsize=(12, 5))

    # Scatter / Strip plot
    sns.stripplot(x=df['Response'], y=df[col], jitter=True, ax=axes[0])
    axes[0].set_title(f'{col} vs Response (Scatter)')

    # Boxplot
    sns.boxplot(x='Response', y=col, data=df, ax=axes[1])
    axes[1].set_title(f'{col} vs Response (Boxplot)')

plt.tight_layout()
plt.show()
```





6.2 Correlation Analysis

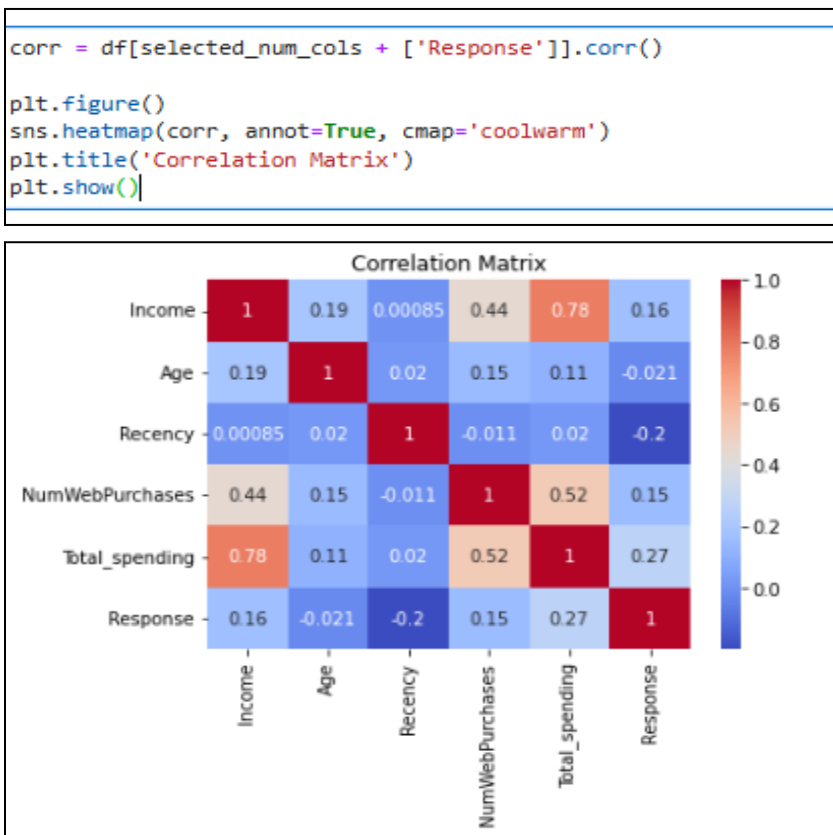
A **correlation heatmap** was generated to examine the relationships between numerical variables and the target variable (**Response**). This visualization helps identify how strongly variables are related to each other and to campaign response behavior.

Interpretation:

- Values close to **+1** indicate a **strong positive correlation**, meaning both variables tend to increase together.

- Values close to **-1** indicate a **strong negative correlation**, meaning one variable increases while the other decreases.
- Values close to **0** indicate a **weak or no linear relationship** between variables.

This analysis helps in identifying the most influential variables that may impact customer response to marketing campaigns.



6.3 Categorical Variables vs Campaign Response

Bar plots were used to analyze **campaign response rates across Education and Marital Status** categories. In addition, **Chi-square tests of independence** were performed to statistically assess whether these categorical variables have a significant relationship with campaign response.

Statistical Results:

- **Education:** p-value = 0.00012226975294505314
- **Marital_Status:** p-value = 1.7384196562604576e-10

Interpretation:

- A **p-value < 0.05** indicates a statistically significant relationship between the categorical variable and campaign response.
- A **p-value ≥ 0.05** indicates no statistically significant relationship.

Based on the results, both **Education** and **Marital Status** show a statistically significant association with campaign response, suggesting that these demographic factors may influence customer engagement with marketing campaigns.

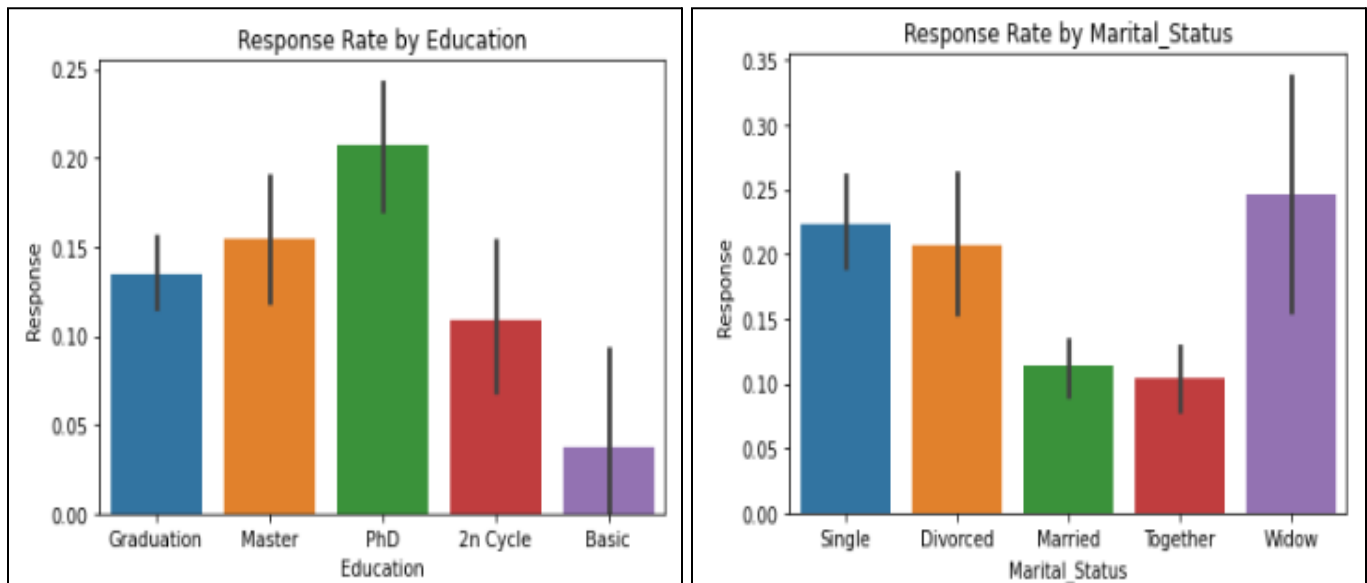
```

selected_cat_cols = ['Education', 'Marital_Status']

for col in selected_cat_cols:
    plt.figure()
    sns.barplot(x=col, y='Response', data=df)
    plt.title(f'Response Rate by {col}')
    plt.xticks
    plt.show()

for col in selected_cat_cols:
    contingency_table = pd.crosstab(df[col], df['Response'])
    chi2, p, dof, expected = chi2_contingency(contingency_table)
    print(f"{col}: p-value = {p}")

```



6.4 Numerical vs Categorical Relationships

To examine how customer demographics influence behavior, **boxplots** were used to compare key numerical variables across **Education** and **Marital Status** categories.

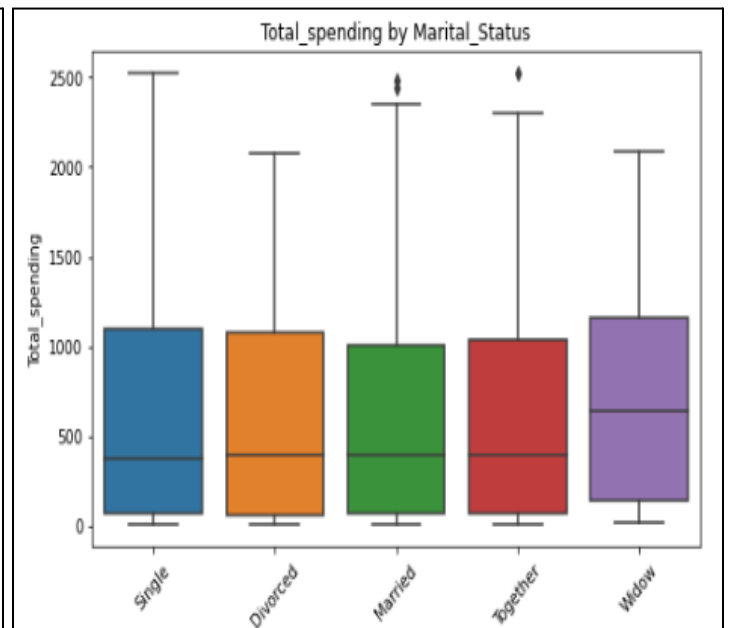
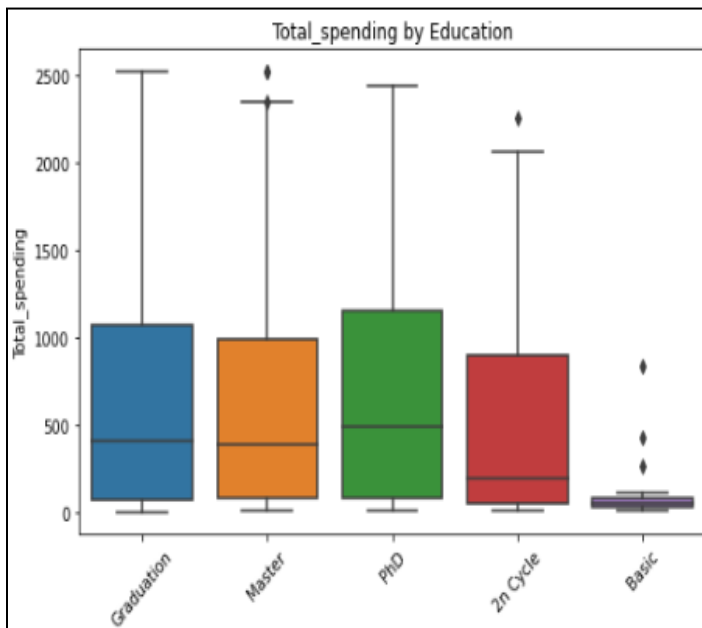
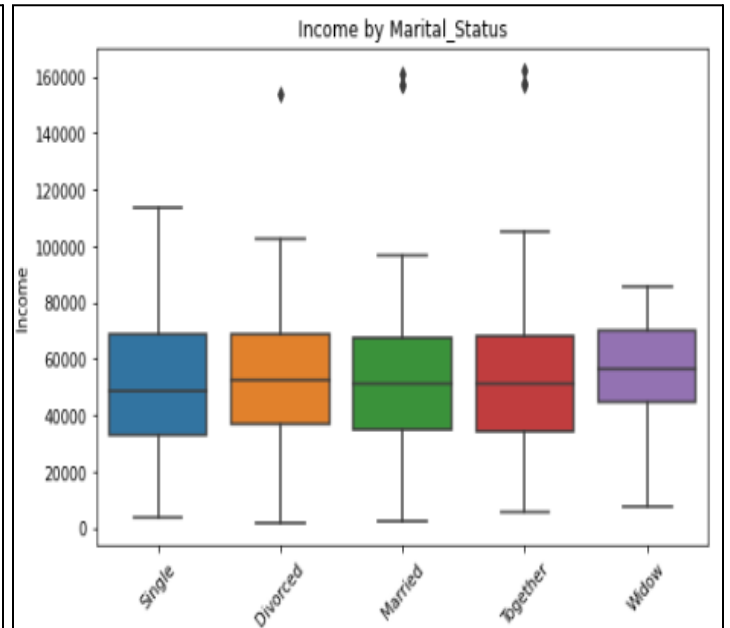
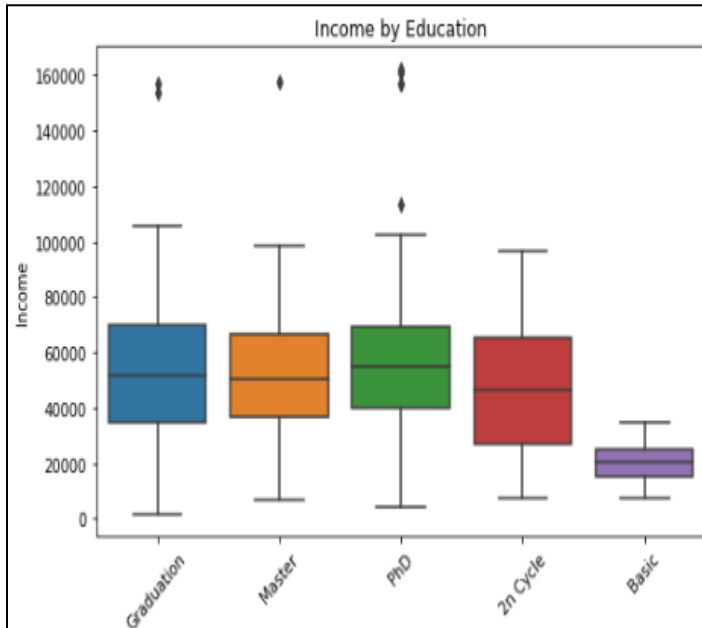
The following relationships were analyzed:

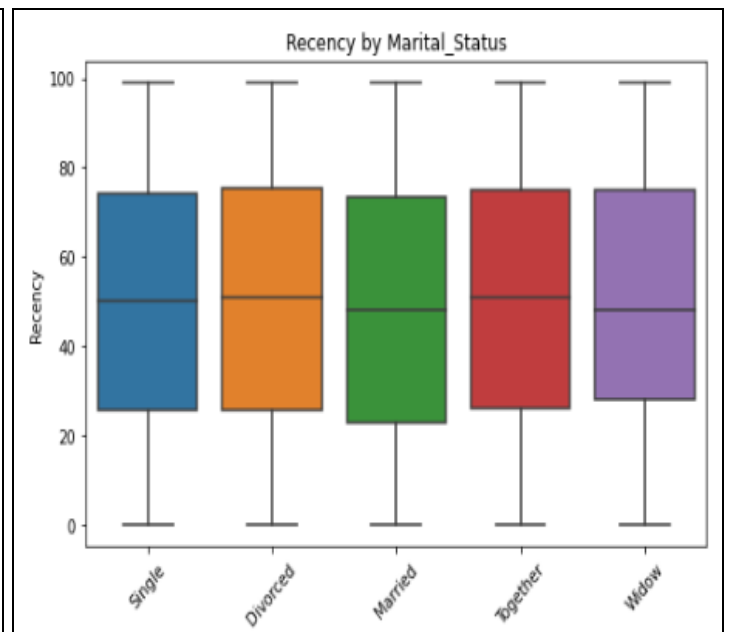
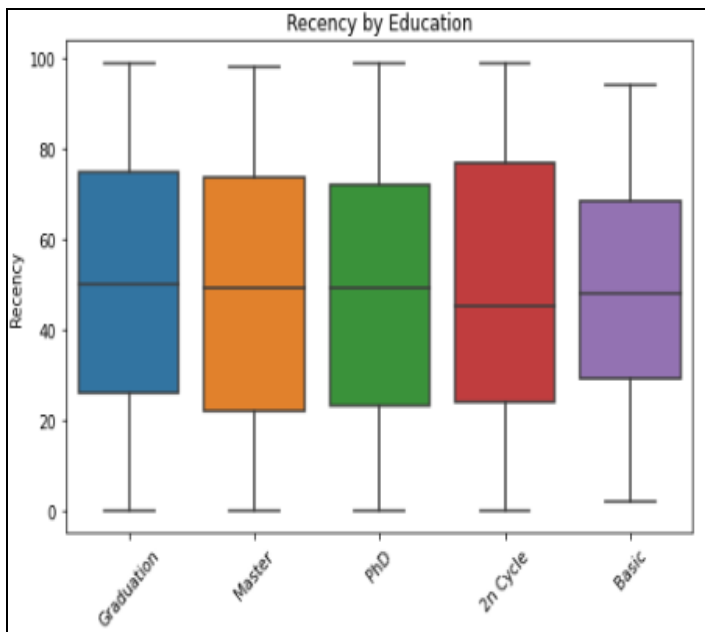
- **Income by Education / Marital Status**
- **Total Spending by Education / Marital Status**
- **Recency by Education / Marital Status**

This analysis provides valuable insights into how different demographic groups vary in terms of **income levels, purchasing behavior, and engagement recency**. It helps identify whether certain customer segments demonstrate distinct financial or behavioral patterns that could be useful for targeted marketing strategies.

```
cat_cols = ['Education', 'Marital_Status']
num_cols = ['Income', 'Total_spending', 'Recency']

for cat in cat_cols:
    for num in num_cols:
        plt.figure(figsize=(8,5))
        sns.boxplot(x=cat, y=num, data=df)
        plt.title(f'{num} by {cat}')
        plt.xticks(rotation=45)
        plt.show()
```





7. Key Observations

Customer Demographics

Analysis of the categorical distributions indicates that the customer base is concentrated within specific education and marital status segments, highlighting the dominant demographic groups represented in the dataset. Additionally, age-related analysis suggests that the majority of customers fall within a mature adult demographic, which may influence purchasing preferences and campaign responsiveness.

Spending Behavior

Income and total spending exhibit substantial variation across customers, indicating heterogeneous purchasing power and engagement levels within the customer base. The presence of multiple outliers in income and spending variables suggests a segment of premium or high-value customers who contribute disproportionately to overall revenue and may warrant separate strategic targeting.

Campaign Response Trends

Bivariate analysis reveals that customers with higher income and greater total spending tend to demonstrate stronger campaign response behavior, suggesting that financially stronger and more engaged customers are more receptive to marketing efforts. Furthermore, recency patterns imply that recently active customers may respond differently to campaigns compared to less active or dormant customers, indicating the importance of customer activity timing in campaign effectiveness.

Statistical Relationships

Correlation analysis identifies meaningful relationships among income, spending variables, and campaign response, suggesting that purchasing behavior metrics may serve as useful predictors in future response modeling. In addition, chi-square test results confirm statistically significant

relationships between campaign response and both education and marital status ($p < 0.05$), indicating that demographic factors play an important role in customer engagement with marketing campaigns.

Business Implications

The findings suggest that customer segmentation strategies based on demographic, financial, and behavioral characteristics could improve marketing campaign targeting and efficiency. Specifically, focusing on high-value customer segments and recently active purchasers may enhance response rates and maximize campaign return on investment.

8. Conclusion

This exploratory data analysis provided valuable insights into customer demographics, purchasing behavior, and campaign response patterns within the marketing campaign dataset. The analysis revealed that customer income, spending behavior, recency of purchases, education level, and marital status all demonstrate meaningful relationships with marketing campaign engagement.

Higher-income and higher-spending customers generally exhibited stronger responsiveness to marketing campaigns, while statistical testing confirmed that demographic variables such as education and marital status have significant associations with campaign response. Additionally, the presence of notable outliers suggests the existence of premium customer segments that may require specialized marketing strategies.

Overall, the findings indicate that customer segmentation based on demographic and behavioral attributes can enhance marketing campaign effectiveness. These insights provide a strong foundation for future predictive modeling and data-driven decision-making aimed at improving customer targeting, engagement, and return on marketing investment. Building on this analysis, predictive machine learning models can be developed to forecast campaign response and identify high-potential customer segments. Variables such as income, recency, spending behavior, and demographic characteristics may serve as key predictors in classification and segmentation models, enabling the organization to further optimize targeting strategies, personalize marketing efforts, and maximize campaign performance.